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Pokémon Random Battle Predictor
Executive Summary

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GITHUB

INTRODUCTION

Pokémon battles are turn-based games in which two players compete with teams of 6 Pokémon. The large number of different Pokémon and different game rulesets allows for enormous variability in gameplay and strategy. One popular game format on battle emulator site Pokémon Showdown ("Showdown") is the "random battle", where players compete with teams of six Pokémon randomly selected at battle start. Ideally, these randomly generated teams should be evenly balanced, so that, going into a battle, both players have nearly equal chances of winning, and thus the winner is indeed "random".

From the Showdown developers' perspective, there are a number of factors and methods to consider in designing this balanced gameplay. Some primary features to consider are Pokémon's stats (e.g., HP, Attack, Defense) and types (e.g., Fire, Psychic, Ground), and players' Elo ratings (within Showdown's system). We aim to investigate whether or not the team generation procedure is balanced. If Showdown's team construction algorithm is balanced, then we expect that comparison of said features should not strongly indicate a player's chance of winning.

DATA COLLECTION

Showdown stores a large amount of information about each match played on the site, including turn-by-turn records of player moves, allowing for a full, turn-by-turn reconstruction of each match from its publicly available "replay" (or "log") JSON file. We saved about 16,000 Showdown match replays, and then employed a custom-made parser (`<class Battle>` in `tools/battle.py`) to extract and organize match information from these JSONs.

The Team Computation Hurdle

In each random battle, players cannot see their opponent's Pokémon until they are individually fielded; additionally, the replay logs do not record these not-yet-fielded Pokémon, even past the battle's end. In order to learn the complete team rosters in each battle—including those Pokémon not fielded—it was necessary to locate the pseudorandom strings used as seeds in generating players' random teams. These seeds were then fed back into Showdown's server source-code to re-generate and store these complete rosters.

FEATURE ENGINEERING

Having two teams of 6 Pokémon, each with their own stats and characteristics, gave us many potential features to consider, for example:

- **Pokémon Attributes:** Features such as a Pokémon's HP, Attack, Defense, and type(s).
- **Type Diversity:** The number of unique Pokémon types on a team.
- **Pokémon Advantage:** A custom statistic approximating the expected difference in total damage that two opposing Pokémon would exchange in a one-on-one match.
- **Total Advantage:** The cumulative sum of the Advantages for all 36 possible pairs of opposing Pokémon from each team.

While Total Advantage was our most impactful feature, it has known drawbacks, such as failing to consider the specific moves or abilities for each Pokémon. In an attempt to address this we introduced the ***Stat-Booster Differential***, which records the difference in the number of Pokémon on each team that know a stat-boosting move, or have a stat-boosting ability.

MODEL SELECTION

For a "baseline" model we used a single-feature predictor based on players' Elo rating differences, since the Elo system maps this difference to estimated probabilities of each player winning. Beyond this baseline, we considered Logistic Regression models both with and without intercept terms (the "biased" and "unbiased" models, respectively), the standard Decision Tree and Random Forest models, and the boosting models HGBoost and XGBoost.

We trained these models on the features which optimized their cross-validation accuracy: for the unbiased Logistic Regression, those features were Elo differential, Total Advantage, and Stat-Booster Differential. The remaining models trained on all of those features plus type diversity differential and total stat differential.

Accuracy was chosen over other metrics for its ease of interpretation and because our application does not require special precautions against “false positives and negatives”, such as applications in, e.g., medical diagnosis. Additionally, we wanted to select a model that was well-calibrated: since Player 1 lost in 52% of the matches, we wanted a model which would predict that Player 1 would lose approximately 52% of the time.

On our training data, the baseline- and unbiased Logistic Regression models were 52.1% and 53.1% accurate, and both were well-calibrated. The remaining models had accuracies all close to the former two, but were poorly calibrated, leading us to select the unbiased logistic regression as our final model.

CONCLUSION AND NEXT STEPS

Conclusion

After selecting the unbiased Logistic Regression as our final model and trialing it on our test data, our model achieved an accuracy of 53.6%. As this is not much better than a coin flip's accuracy of 50%, we conclude that the level and move-set scaling performed by Pokémon Showdown for Gen-9 random battles is sufficiently balanced to maintain interesting and unpredictable gameplay.

Next Steps

There remains potential to create a predictive model based on turn-by-turn analysis. Instead of relying on statistics determined pre-game, this model could estimate player win probabilities as a function of the game “state”, updating its predictions after every turn. In addition, the Advantage metric can be further refined to better account for Pokémon's special abilities and held items.